The Limits of Targeted Hiring Subsidies: Evidence from the Work Opportunity Tax Credit

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How can governments encourage employment of disadvantaged groups?

Longstanding concern about labor market prospects of disadvantaged groups

- 25pp gap in employment between high-educated and low-educated (NCES '22)
- 55% of formerly incarcerated have earnings in 1st year post-release (Looney & Turner '18)

Labor supply policies to improve the skills of workers and increase job-seeking

- E.g., job training and education programs, EITC

Labor demand policies to hire more disadvantaged workers or increase wages

- E.g., minimum wages, **employer-side wage subsidies**

This paper: the Work Opportunity Tax Credit (WOTC)

Research questions:

- Do wage subsidies increase hiring of targeted groups?
- What is the pass-through of subsidies onto earnings?
- Do subsidies reduce use of government benefit programs?
- Do capped subsidies encourage "churning"?
- What mechanisms mediate the relationship between subsidies and employment?

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Study largest employer-side wage subsidy program in US: **Work Opportunity Tax Credit**

- 2022: subsidized hiring/wages of over 2.5 million workers (over 3% of all hires)
- Targeted to disadvantaged populations (e.g., public assistance, felony conviction)
- Subsidy to firm for new hires at up to 40% of earnings, cap of $2,400 in subsidy/worker
- Reauthorization under debate in Congress, incl. bipartisan bill to substantially expand go

This paper: the Work Opportunity Tax Credit

Data: Linked administrative data from Wisconsin (2000-2019)

- Earnings, SNAP, UI benefits, TANF, courts, DOC, WOTC micro-data

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Four empirical strategies: Examine employment, tenure, earnings, public benefits, crime

- Event study/DiD around expansion of eligibility
- Age-based discontinuities for eligibility
- Firm-level matched diff-in-diff around expansion of eligibility
- Event study/DiD around reduction in application costs

Preview of findings

Consistent and precise zeros across all approaches

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- No effects on employment

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But millions of workers' wages are subsidized by WOTC

- Inframarginal: subsidies go to firms for hiring workers they would have hired anyway

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- Benefits are heavily concentrated in small set of firms

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Why?

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Why? Two general frictions that limit employer-side targeting

- **External**: firms rarely collect data on elig. at hiring (perceived legal risk, deter applicants)
- **Internal**: hiring/retention decision-makers don't internalize subsidy (agency, info frictions)

Related literature

Employer-side wage subsidies: older US programs with survey data; recent US work bundled with training, smaller-scale state credits or experiments e.g., Perloff and Wachter 1979; Bishop and Montgomery 1993; Katz 1996; Gelber et al. 2016; Grijalva and Neumark 2017; Cullen et al. 2023; Bushway and Pickett 2024; outside the US: Cahuc et al. 2019; Saez et al. 2019; Fenizia et al. 2025

→ *This paper*: precise quasi-experimental estimates for the largest US program, at a policy-relevant scale; highlight new mechanisms

Frictions in hiring: perceived legal constraints and delegated hiring e.g., Acemoglu and Angrist 2001; Autor et al. 2006; Agan and Starr 2018; Aizawa et al. 2024; Cowgill and Perkowski 2024; Aleksenko and Kohlhepp 2024

→ *This paper*: these frictions can undo targeted subsidies

WOTC: descriptive or weaker causal designs Hamersma 2008; Hamersma and Heinrich 2008; Heaton 2012; Hunt et al. 2018

→ *This paper*: quasi-experimental approaches with linked administrative data; a less favorable appraisal

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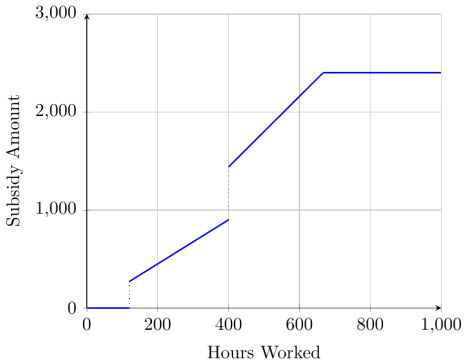
Work Opportunity Tax Credit: Background

- Began in 1997 after end of earlier program (Targeted Job Tax Credit)
- Tax credit *to firms* for 25-40% of earnings, depending on hours; cap of $2400 subsidy

Target groups (roughly):

- **SNAP recipients**
- TANF recipients
- 1 yr post felony conviction/release
- Long-term unemployed
- Designated Community Residents
- SSI recipients
- Veterans
- ARRA Disconnected Youth

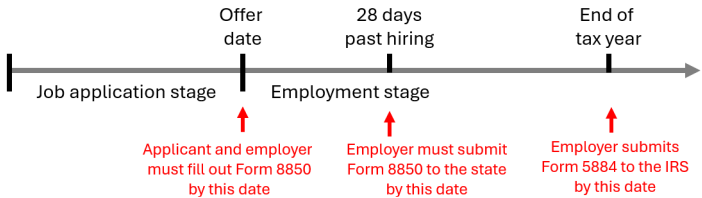
Ex: $9/hr worker (4 mo FT to maximize)



Steps to receive WOTC

Firms must:

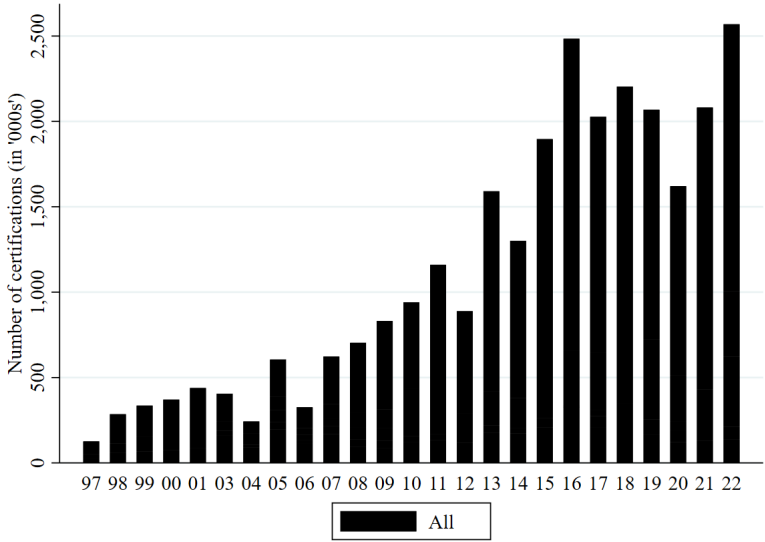
1. Fill out form 8850 "on or before the day a job was offered to the applicant" Examples BtB
2. Verify eligibility with state (State Workforce Agency) tax form rejection
3. Then claim on federal taxes, can carry forward tax form



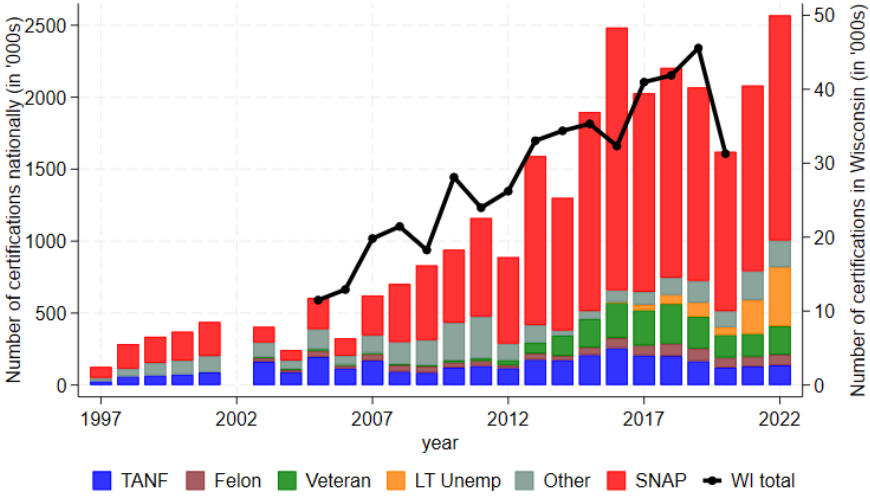
Firm can only get subsidy once per worker, but no firm-level limit on overall certifications

- Other firms can receive subsidy for same worker in later employment stints
- Eligibility only matters at time of hire; subsidy continues even if changes while employed

Massive program expansion over the past 15 years

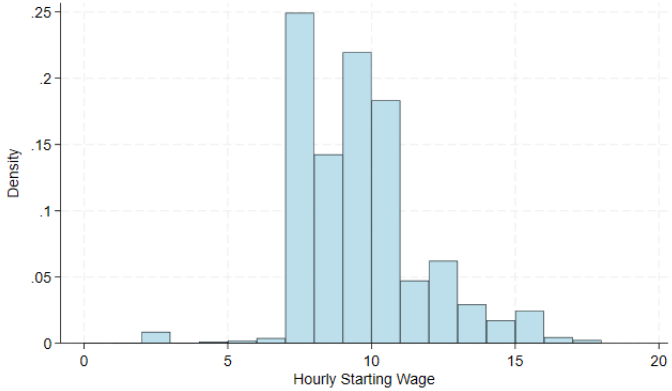


SNAP is by far the largest target group



WOTC starting hourly wage distribution back

- In our sample, typically earn \$7.25-\$11, median is \$9 per hour
- Top occupations: retail, employment services, food prep/serving



Data

Linked administrative data from various government agencies from Wisconsin (pop: 6 million)

- Follows 13.3 million individuals from 1988 to 2020
- Includes:
  - SNAP benefits
  - Quarterly employment/earnings (all workers covered by UI)
  - WOTC applications and certifications: 800,000 applications from 2005-2019
  - TANF benefits
  - Court and incarceration records
  - Unemployment insurance benefits

Other data sources:

- Data on application process for firms that benefit from WOTC
- Surveys of hiring decision-makers and of supervisors at WOTC-using firms
- FOIAed WOTC micro-data from Wisconsin and 7 other states (2018-2020)

Additional facts

Summary statistics table

Wage distribution graph

Firm-level stats more firm stats distribution of benefits

Application and success rates application rate

Payroll share payroll share

Share of individuals in poverty who are WOTC-eligible CPS

Why employer-side subsidies? Incidence with a binding minimum wage graph

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Effect of expansion of WOTC eligibility for SNAP recipients eligible

Prior to 2007: Indiv. 18-24 who received SNAP in 3 of past 5 months eligible for WOTC

- In 2007q1, this constituted around 30,000 eligible individuals

Policy change in 2007: Expanded to include SNAP beneficiaries aged 25-39

- In 2007q1, this constituted around 60,000 newly eligible individuals

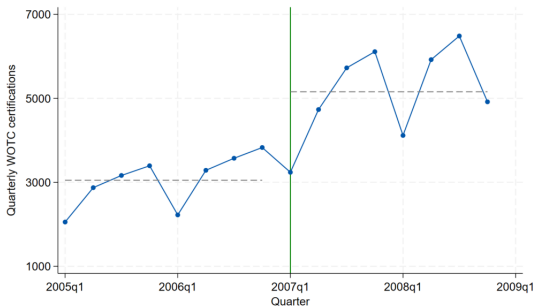
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Diff-in-diff/event study before (2005-06) vs after (2007-08) policy change

Main approach: Compare newly treated to just untreated

- Treated group: Ages 38-39 and receive SNAP
- Untreated group: Ages 42-43 and receive SNAP
- Exclude age 40-41 because they are partially treated in 2007-08

Estimating equation for SNAP beneficiary i in quarter t:

$$Y_{i,t} = \sum_{q=-6}^{8} \beta_q Treated_{a(i,t)} \times \mathbb{1}\{t = q\} + X_i\phi + \eta_t + \gamma_a + \varepsilon_{it} \quad (1)$$

- $Y_{i,t}$: new hire, employment, earnings, public benefit use, criminal conviction
- Year-quarter FEs (η_t) and age FEs (γ_a)

Diff-in-diff/event study before (2005-06) vs after (2007-08) policy change

Secondary approach: Compare newly treated to always treated

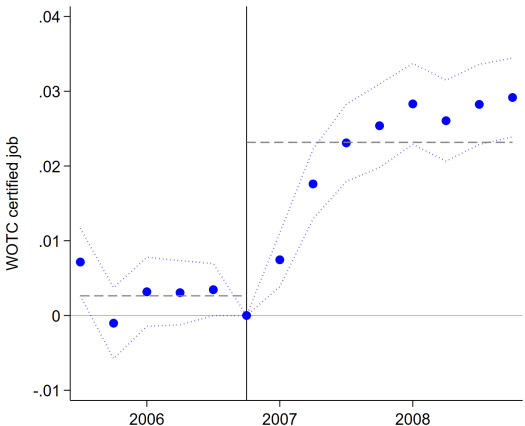
- Newly treated group: Ages 25-26 and receive SNAP
- Always treated group: Ages 23-24 and receive SNAP

Paper presents results for each approach separately, as well as pooled estimates

- Results are the same; show the pooled estimates today

Employed in a WOTC-certified job

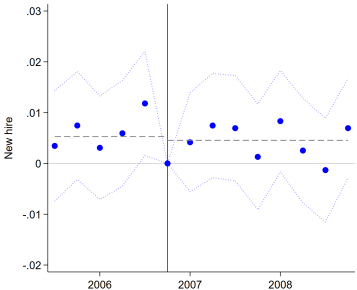
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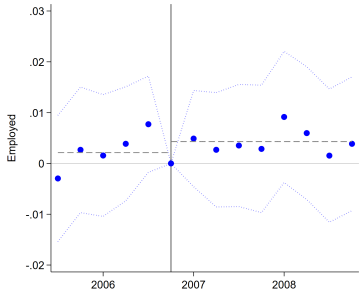


	(1) WOTC job
Treat x post	0.0208*** (0.0015)
Dep var mean	0.0138
Observations	527,241

Effects on **employment**: did demand for targeted workers shift out?

Effects on **employment**: did demand for targeted workers shift out?





- Precise zero: no detectable effect on employment or transition into new jobs for target group

	New hire	Employed
Treat x post	-0.0008	0.0021
	(0.0022)	(0.0039)
Dep var mean	0.157	0.549
Observations	527,241	527,241

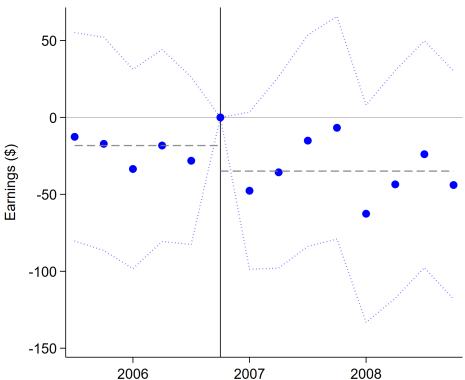
Elasticities | Unemployed sample | Triple diff

Subsidy may still pass-through to worker earnings

- Firms may retain workers longer for the subsidy; subsidized workers may get higher wages

Subsidy may still pass-through to worker earnings

- Firms may retain workers longer for the subsidy; subsidized workers may get higher wages
- Precise null effect on earnings so neither appears to be the case



	Earnings
Treat x post	-16.72
	(20.83)
Dep var mean	2249.42
Observations	527,241

unemployed sample public benefits, crime

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Age discontinuities in eligibility

Age cut-offs for WOTC eligibility under SNAP:

- **Pre-2007**: Ages 18-24
- **Post-2007**: Ages 18-39

This produces sharp discontinuities around the age cut-off in eligibility

- RDD around age 25 for 1998-2006; RDD around age 40 for 2007-2019

Advantages of this approach:

- Can estimate the treatment effect over time
- Better statistical power for heterogeneity analysis

eligible composition check

Regression discontinuity effect of WOTC eligibility for SNAP recipients

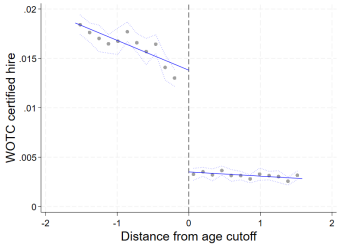
Estimating equation for RDD for i in quarter t for cutoff A:

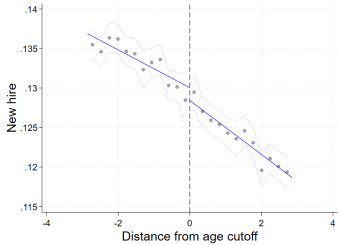
$$Y_{i,t} = \gamma_0 + \tau 1\{age_{i,t} \geq A\} + f_1(age_{i,t}) + f_2(age_{i,t}) 1\{age_{i,t} \geq A\} + \epsilon_{i,t} \quad (2)$$

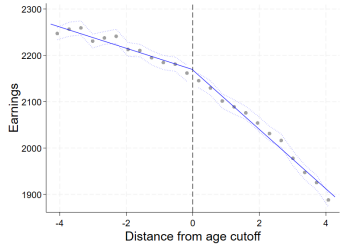
- A is equal to 25 for years < 2007; 40 for years ≥ 2007
- $Y_{i,t}$: new hire, employment, earnings
- Follow Calonico et al (2014) for selection of bandwidth; use triangular kernel function; cluster at individual-level

eligible composition check density

No detectable difference in employment or earnings







	WOTC certified hire	New hire	Earnings
RD_Estimate	-0.0097***	-0.0007	-3.1037
	(0.0004)	(0.0010)	(12.0372)
Dep var mean	0.003	0.124	2032.539
Bandwidth	1.585	2.858	4.272
Effective Obs	1255960	2782835	4174431

split by year | public benefits | crime | placebo | composition | first stage | balance | Do firms know the age cutoff?

Additional RD analysis

Program has expanded, perhaps it took time for firms to adjust HR policies?

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- No differences in (lack of) treatment effect throughout the sample period go

No effect overall, but perhaps some firms are responsive?

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- No differences in (lack of) treatment effect throughout the sample period go

No effect overall, but perhaps some firms are responsive? No effect on being hired at: go

- Top 50 WOTC using firms
- Employment services firms (NAICS 5613), the biggest users of WOTC
- Firms highly attentive to WOTC (certified for high share of eligible hires)
- Firms in more competitive sectors (4 digit NAICS)
- Firms that explicitly ask WOTC questions on job applications

individual heterogeneity balance Do firms know the age cutoff? placebo composition first stage

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Combining estimates, additional approaches

We combine estimates of inframarginality from the DD and RD strategies

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- Estimator is the ratio of treatment effect on new hires divided by treatment effect on WOTC certifications

Combining estimates, additional approaches

We combine estimates of inframarginality from the DD and RD strategies

- Estimator is the ratio of treatment effect on new hires divided by treatment effect on WOTC certifications
- Point estimate: 96.2% of hires are inframarginal; cannot reject null that all hires are inframarginal ($p = 0.72$)

Paper also evaluates similar questions using two other approaches

- (3) Firm-level matched DD: WOTC-using firms vs. matched non-users, '07 expansion go
- (4) DD around 2017 switch to electronic filing (which lowered application costs) go

Who benefits from WOTC?

- Over half of certifications in Wisconsin (2005–2019) went to 48 firms, which accounted for 9% of all hires go

firm adoption design

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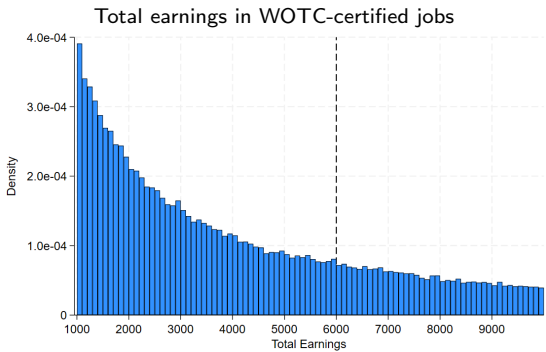
Do firms "churn" workers once the subsidy is exhausted?

Subsidy is capped at $2,400 per worker, typically reached at $6,000 in earnings

- **Churning**: dismiss workers at the cap to hire new subsidized workers
  $\Rightarrow$ bunching just above $6,000
- **Hazard**: cost of employment jumps at exhaustion
  $\Rightarrow$ separation hazard rises after $6,000

Do firms "churn" workers once the subsidy is exhausted?

No bunching (density test $p = 0.86$)

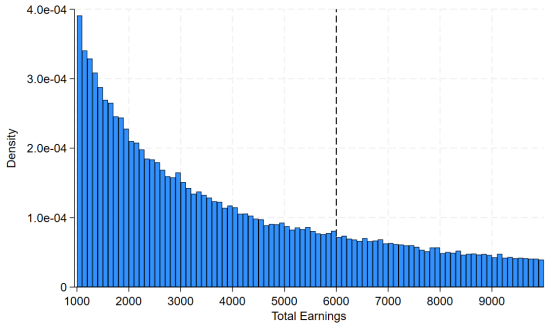


hazard table | by firm type

Do firms "churn" workers once the subsidy is exhausted?

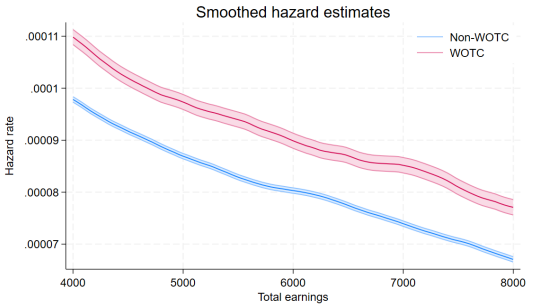
No bunching (density test $p = 0.86$) ... and no change in separation hazard at \$6,000

Total earnings in WOTC-certified jobs



hazard table | by firm type

Hazard of separation, WOTC vs. non-WOTC hires



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Why don't firms respond? (1) External constraints on observing eligibility

Eligibility hinges on potentially sensitive characteristics (e.g., SNAP receipt, age)

- Not observable from outside sources (except felony convictions, $<4\%$ of certifications)
- Firms must rely on applicants' self-reports, and may be reluctant to ask

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Do firms collect eligibility information before hiring? details

- FOIA data with names of all WOTC-using firms in Wisconsin, 2018–2020
- Sample 179 firms with probability proportional to their number of WOTC applications
- Locate an entry-level job application for 138 (77%), jointly 53% of WI certifications
- Record whether the application asks about WOTC eligibility, and who collects the answers

(1) External constraints: firms rarely know eligibility when hiring

Job applications table

- Only **22%** ask about WOTC eligibility on the application
- Of these, **60%** send answers to a third party (tax servicer/payroll), often not shared with the employer example
- Most firms instead collect it **after** the hiring decision (onboarding) National Express McMillan
- $\Rightarrow$ Hiring decision-makers likely know eligibility in fewer than 1 in 10 cases

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Why not ask?

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Why not ask? Survey of hiring decision-makers at non-screening WOTC firms (N=65) details workers

- 60%: fear of discrimination lawsuits (e.g., favoring under-40s) EEOC opinion
- 42%: hard to explain to applicants why we are asking
- 35%: don't want to scare people off from applying

Would firms respond if they could observe eligibility?

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Test among firms that screen for WOTC on the application **and** collect it themselves

- Link job application data to the administrative data
- Add the next 60 largest WOTC filers for power (sample covers 62.6% of WI WOTC applications)
- Test for employment and hiring responses among the set of firms that we know screen for WOTC eligiblity on their applications

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Still no effect on employment or hiring at these firms RD DD

- Points to the presence of other frictions that limit the response to targeted subsidies
- $\Rightarrow$ Removing external constraints alone may not be enough

Why don't firms respond? (2) Internal organizational frictions

Hypothesis: hiring and retention decisions are delegated to managers who:

- do not directly benefit from the subsidy (agency issues)
- may not know which applicants are eligible (internal information frictions)

Two pieces of suggestive evidence on this channel:

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- No missing mass below either notch, even at screening firms hours hazard

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(2) Surveys of hiring decision-makers and supervisors at WOTC firms details

- *Agency*: "only the company benefits" (47% of hiring decision-makers, 85% of supervisors)
- *Information*: 71% of supervisors say managers don't know who brings in WOTC money

skip to conclusions

Additional hypotheses

Could other mechanisms explain the lack of response?

- **Up-front costs** of integrating WOTC into hiring systems? No effect even at screening firms among top 50 WOTC users go
- **Attentiveness** to WOTC? No effect even at screening firms that are highly attentive, as measured by applying for a high fraction of eligible hires go
- **Stigma** offsetting the subsidy? No drop in hiring of SNAP recipients aged 40+ after adoption; no response to e-filing adoption e-filing

Substitutability Audit study

Conclusions

We study labor market effects of wage subsidies to firms for hiring disadvantaged workers

- Across four designs, similar and precise null effects, suggesting firms receive subsidies for workers they would hire anyway
- Benefits are concentrated: over half of certifications go to 48 firms with 9% of hires
- Highlight general frictions that limit employer-side targeting: perceived legal risk and organizational frictions

Policy importance: reauthorization under debate, incl. bipartisan bill that would more than triple program cost go

- Possible reforms: enforce determining eligibility before job offer; make it easier to identify eligible candidates (e.g., pre-certification)
- But even with reforms, funds may be more effectively deployed to other support for disadvantaged workers

Thank you!

Additional slides

Summary statistics

return

	WOTC applicants	WOTC certified	SNAP sample, 18-39
Panel A: Demographic characteristics			
Age	29.1	28.2	28.7
Male	0.41	0.36	0.35
White	0.59	0.55	0.63
Black	0.37	0.43	0.32
Hispanic	0.10	0.10	0.11
Born in WI	0.62	0.64	0.64
SNAP beneficiary in quarter	0.63	0.77	0.88
Quarterly SNAP benefits, cond. ($)	945.5	1042.6	1012.1
TANF beneficiary in quarter	0.064	0.100	0.064
UI beneficiary in quarter	0.083	0.083	0.065
Felony conviction in quarter	0.012	0.012	0.014
Panel B: Job stints			
Num. quarters with firm	3.50	3.42	3.54
With firm over 3 quarters	0.25	0.24	0.25
Total earnings at job (median)	2771.0	2578.9	2476.6
Total earnings at job (mean)	13563.9	12530.4	14470.1
Starting wage	9.15	9.02	—
Industry: employment services	0.27	0.27	0.22
Industry: restaurants	0.14	0.14	0.13
Industry: retail	0.31	0.31	0.14
Median quarterly employees of firm	1352.8	1304	276.5
Median quarterly payroll of firm	4974766.5	4836772	1176011.9
Applied for WOTC	1	1	0.21
Certified for WOTC	0.55	1	0.17
Total hours: 0-120	0.33	0.35	
Total hours: 120-400	0.23	0.24	
Total hours: 400+	0.43	0.42	
Total hours at job, median	286.7	266.9	
WOTC subsidy ($)	608.5	1097.9	
Firm received max WOTC subsidy	0.18	0.33	
Pre-certified	0.012	0.015	
Unique individual-quarters	640818	354829	10985951
Unique individuals	415828	234557	867916

Statistics on firms using WOTC back

- Overall, an application is filed for between 10-17% of eligible hires (unconditional on being an adopting firm), with a gradual increase over time graph
- Firms are large: median quarterly payroll: $4.8 million, median employment: 1,304
- After adoption, file WOTC applications for 11.5% of all hires
- Apply for 37.2% of eligible hires, increases with experience
- 12% of all hires are eligible for WOTC
- 81% of firms existed for at least one year prior to first usage of WOTC
- Application success rate varies from between 50 to 65% graph

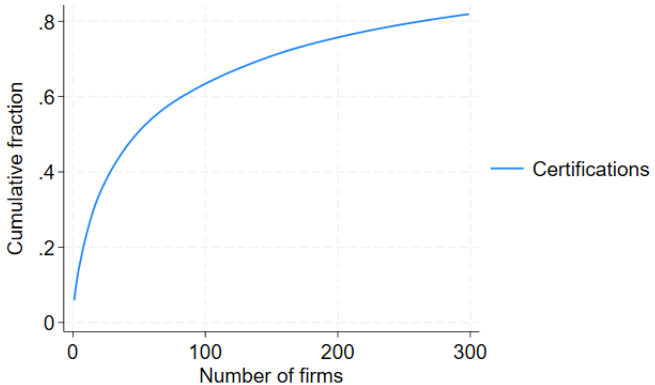
Who benefits from WOTC? back

Benefits appear to accrue to firms. So which firms are benefiting?

Who benefits from WOTC? back

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Over half of all certifications in Wisconsin from 2005-2019 went to **48 firms**

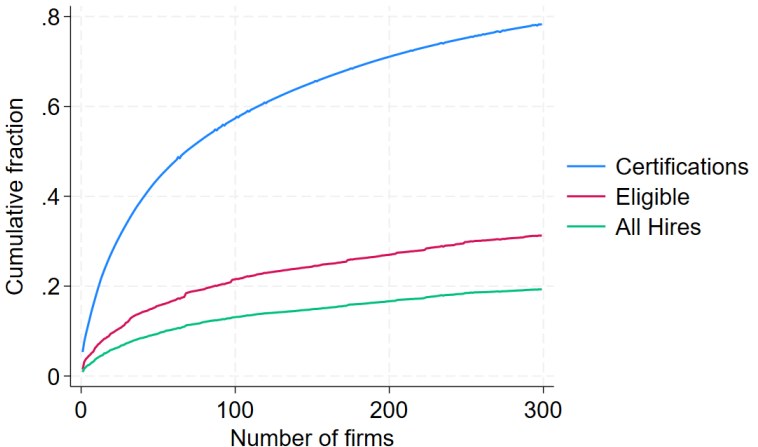


Who benefits from WOTC? back

Benefits appear to accrue to firms. So which firms are benefiting?

Over half of all certifications in Wisconsin from 2005-2019 went to **48 firms**

- In comparison, these 48 firms accounted for **9% of all hires** over the period



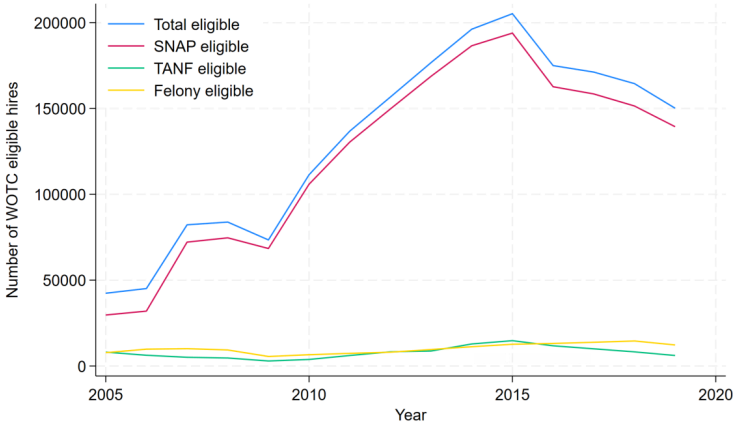
Change in WOTC application rates over time back

- Application success has been steady over time
- Steady increase in rate of applying for eligible workers



Why did WOTC expand so much? Ans: increase in eligibility rates of hires

- Wisconsin data on eligibility for SNAP, TANF, felonies
- Expansion in WOTC certifications coincides with the increase in WOTC-eligible new hires (mostly driven by SNAP)

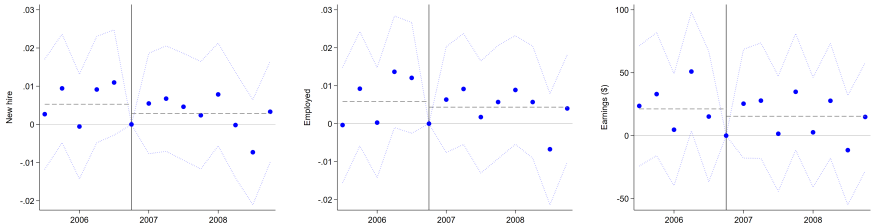


DD effect of WOTC eligibility, unemployed sample back

DD effect of WOTC eligibility, unemployed sample

back

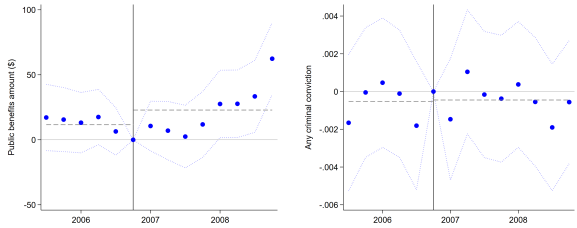
- Precise zero: rule out hiring effect >0.33pp at 95% confidence level



	New hire	Employed
Treat x post	-0.0026	-0.0017
	(0.0030)	(0.0032)
Dep var mean	0.1719	0.2046
Observations	304,498	304,498

DD effects on public benefit use (SNAP, TANF, UI), crime

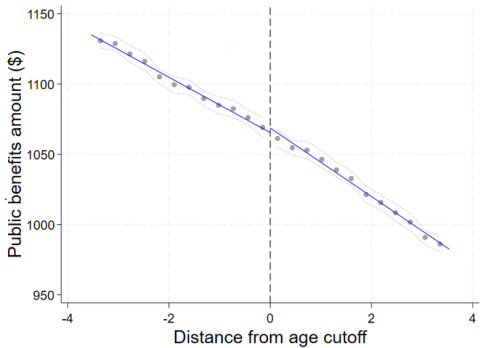
back



	Public benefit (amt)	Any conviction
Treat x post	12.58	0.0001
	(8.08)	(0.0007)
Dep var mean	908.33	0.0153
Observations	527,241	527,241

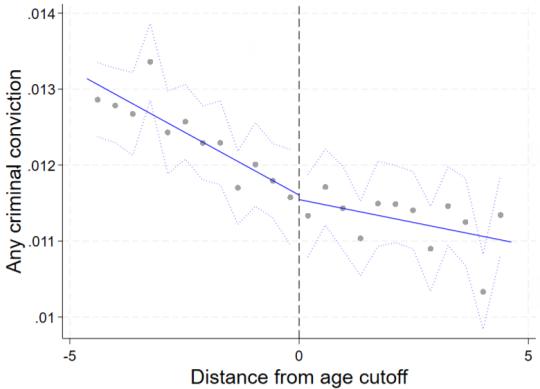
No effects of WOTC eligibility on **public benefit** use (SNAP, TANF, UI)

back



	Public Benefit
RD Estimate	0.8321
	(4.2601)
Dep var mean	1025.865
Bandwidth	3.539
Effective Observations	3438705

No effects of WOTC eligibility on **criminal activity** back



Reduced cost of access due to e-filing

Prior to 2017, employers mailed/faxed application forms for each worker within 28 days of hire

In June 2017, electronic filing implemented in WI

Strong evidence in other settings that application costs deter applications/take-up

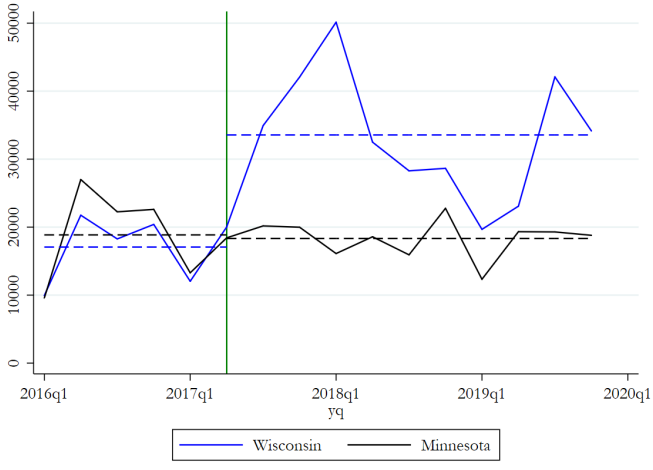
- E-filing personal income taxes and EITC take-up (Kopczuk and Pop-Eleches 2007)
- Social Security Disability Insurance, esp. for most vulnerable (Deshpande and Li 2019)

Here, employers apply for benefits, not workers

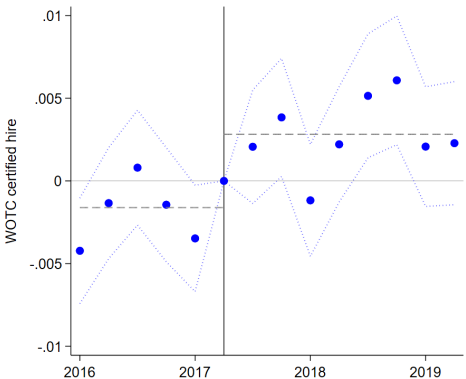
- Larger firms may find it less costly to apply (economies of scale)
- Lowering application costs may increase applications from smaller firms, whose responses to subsidies may differ from large firms

Large increase in applications following e-filing

- Clear expansion, as seen relative to Minnesota by category
- Larger response from low-WOTC usage firms by firm size by application history

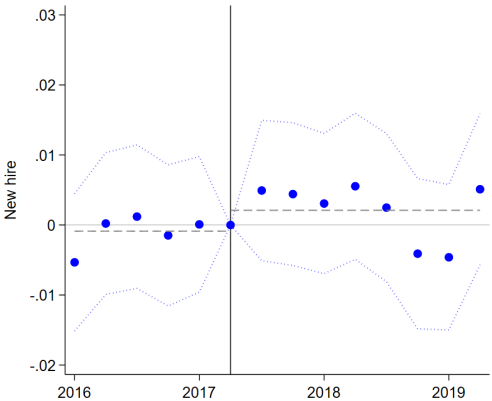


Effect of 2017 adoption of e-filing on **WOTC certifications**



	WOTC cert. hire
Treat x post	0.0044***
	(0.0007)
Dep var mean	0.0041
Observations	391,985

Effect of 2017 adoption of e-filing on **new hires**



	New hire
Treat x post	0.0031
	(0.0023)
Dep var mean	0.1032
Observations	391,985

WOTC reauthorization / expansion bill back

Work Opportunity Tax Credit requires regular reauthorization by Congress every ∼5 years

- Most recent authorization ran through December 2025; reauthorization currently under debate

Bipartisan bill to renew and expand the program (Improve and Enhance the WOTC Act; 9 Senate co-sponsors)

- Increases wage subsidy to 50% instead of 40%
- Increases earnings cap from $6,000 to $12,000 for most workers
- Would remove age caps for SNAP
- Would more than triple the program's annual cost (EY, 2025)

Initial application (Form 8850) Back

Form **8850** (Rev. March 2016)
Department of the Treasury
Internal Revenue Service

Pre-Screening Notice and Certification Request for the Work Opportunity Credit

▶ **Information about Form 8850 and its separate instructions is at *www.irs.gov/form8850*.**

OMB No. 1545-1500

Job applicant: Fill in the lines below and check any boxes that apply. Complete only this side.

Your name _____ Social security number ▶ _____

Street address where you live _____

City or town, state, and ZIP code _____

County _____ Telephone number _____

If you are under age 40, enter your date of birth (month, day, year) _____

1 ☐ Check here if you received a conditional certification from the state workforce agency (SWA) or a participating local agency for the work opportunity credit.

2 ☐ Check here if **any** of the following statements apply to you.
- I am a member of a family that has received assistance from Temporary Assistance for Needy Families (TANF) for any 9 months during the past 18 months.
- I am a veteran and a member of a family that received Supplemental Nutrition Assistance Program (SNAP) benefits (food stamps) for at least a 3-month period during the past 15 months.
- I was referred here by a rehabilitation agency approved by the state, an employment network under the Ticket to Work program, or the Department of Veterans Affairs.
- I am at least age 18 but **not** age 40 or older and I am a member of a family that:
  a. Received SNAP benefits (food stamps) for the past 6 months; **or**
  b. Received SNAP benefits (food stamps) for at least 3 of the past 5 months, **but** is no longer eligible to receive them.
- During the past year, I was convicted of a felony or released from prison for a felony.
- I received supplemental security income (SSI) benefits for any month ending during the past 60 days.
- I am a veteran and I was unemployed for a period or periods totaling at least 4 weeks but less than 6 months during the past year.

3 ☐ Check here if you are a veteran and you were unemployed for a period or periods totaling at least 6 months during the past year.

4 ☐ Check here if you are a veteran entitled to compensation for a service-connected disability and you were discharged or released from active duty in the U.S. Armed Forces during the past year.

5 ☐ Check here if you are a veteran entitled to compensation for a service-connected disability and you were unemployed for a period or periods totaling at least 6 months during the past year.

6 ☐ Check here if you are a member of a family that:
- Received TANF payments for at least the past 18 months; **or**
- Received TANF payments for any 18 months beginning after August 5, 1997, **and** the earliest 18-month period beginning after August 5, 1997, ended during the past 2 years; **or**
- Stopped being eligible for TANF payments during the past 2 years because federal or state law limited the maximum time those payments could be made.

7 ☐ Check here if you are in a period of unemployment that is at least 27 consecutive weeks and for all or part of that period you received unemployment compensation.

Signature—All Applicants Must Sign

Under penalties of perjury, I declare that I gave the above information to the employer on or before the day I was offered a job, and it is, to the best of my knowledge, true, correct, and complete.

For Employer's Use Only

Employer's name _____ Telephone no. _____ EIN ▶ _____

Street address _____

City or town, state, and ZIP code _____

Person to contact, if different from above _____ Telephone no. _____

Street address _____

City or town, state, and ZIP code _____

If, based on the individual's age and home address, he or she is a member of group 4 or 6 (as described under *Members of Targeted Groups* in the separate instructions), enter that group number (4 or 6) . . . . . ▶ _____

Date applicant:
Gave information _____ Was offered job _____ Was hired _____ Started job _____

Under penalties of perjury, I declare that the applicant provided the information on this form on or before the day a job was offered to the applicant and that the information I have furnished is, to the best of my knowledge, true, correct, and complete. Based on the information the job applicant furnished on page 1, I believe the individual is a member of a targeted group. I hereby request a certification that the individual is a member of a targeted group.

Employer's signature ▶ **Title** **Date**

Privacy Act and Paperwork Reduction Act Notice

Section references are to the Internal Revenue Code.

Section 51(d)(13) permits a prospective employer to request the applicant to complete this form and give it to the prospective employer. The information will be used by the employer to complete the employer's federal tax return. Completion of this form is voluntary and may assist members of targeted groups in securing employment. Routine uses of this form include giving it to the state workforce agency (SWA), which will contact appropriate sources to confirm that the applicant is a member of a targeted group. This form may also be given to the Internal Revenue Service for administration of the Internal Revenue laws, to the Department of Justice for civil and criminal litigation, to the Department of Labor for oversight of the certifications performed by the SWA, and to cities, states, and the District of Columbia for use in administering their tax laws. We may also disclose this information to other countries under a tax treaty, to federal and state agencies to enforce federal nontax criminal laws, or to federal law enforcement and intelligence agencies to combat terrorism.

You are not required to provide the information requested on a form that is subject to the Paperwork Reduction Act unless the form displays a valid OMB control number. Books or records relating to a form or its instructions must be retained as long as their contents may become material in the administration of any Internal Revenue law. Generally, tax returns and return information are confidential, as required by section 6103.

The time needed to complete and file this form will vary depending on individual circumstances. The estimated average time is:

Recordkeeping . . 6 hr., 27 min.
Learning about the law or the form . . . . . 24 min.
Preparing and sending this form to the SWA . . . . . 31 min.

If you have comments concerning the accuracy of these time estimates or suggestions for making this form simpler, we would be happy to hear from you. You can send us comments from *www.irs.gov/formspubs*. Click on "More Information" and then on "Give us feedback." Or you can send your comments to:

Internal Revenue Service
Tax Forms and Publications
1111 Constitution Ave. NW, IR-6526
Washington, DC 20224

Do not send this form to this address. Instead, see *When and Where To File* in the separate instructions.

Initial application (Form 8850) Back

On penalty of perjury, firms must state their awareness of eligibility prior to hiring, and employee must also affirm that told their status to employer

Signature—All Applicants Must Sign

Under penalties of perjury, I declare that I gave the above information to the employer on or before the day I was offered a job, and it is, to the best of my kn correct, and complete.

Job applicant's signature ▶ **Date**

For Privacy Act and Paperwork Reduction Act Notice, see page 2. Cat. No. 22851L Form **885**

Date applicant:

Gave information _____ Was offered job _____ Was hired _____

Under penalties of perjury, I declare that the applicant provided the information on this form on or before the day a job was o information I have furnished is, to the best of my knowledge, true, correct, and complete. Based on the information the job a believe the individual is a member of a targeted group. I hereby request a certification that the individual is a member of a ta

Employer tax form (form 5584) Back

Form **5884**
(Rev. March 2021)
Department of the Treasury
Internal Revenue Service

Work Opportunity Credit

► **Attach to your tax return.**
► **Go to *www.irs.gov/Form5884* for instructions and the latest information.**

OMB No. 1545-0219

Attachment Sequence No. **884**

Name(s) shown on return

Identifying number

1	Enter on the applicable line below the total qualified first- or second-year wages paid or incurred during the tax year, and multiply by the percentage shown, for services of employees who are certified as members of a targeted group.			
a	Qualified first-year wages of employees who worked for you at least 120 hours but fewer than 400 hours . . .	$ ____ × 25% (0.25)	**1a**	
b	Qualified first-year wages of employees who worked for you at least 400 hours	$ ____ × 40% (0.40)	**1b**	
c	Qualified second-year wages of employees certified as long-term family assistance recipients	$ ____ × 50% (0.50)	**1c**	
2	Add lines 1a, 1b, and 1c. See instructions for the adjustment you must make to your deduction for salaries and wages		**2**	
3	Work opportunity credit from partnerships, S corporations, cooperatives, estates, and trusts (see instructions)		**3**	
4	Add lines 2 and 3. Cooperatives, estates, and trusts, go to line 5. Partnerships and S corporations, stop here and report this amount on Schedule K. All others, stop here and report this amount on Form 3800, Part III, line 4b		**4**	
5	Amount allocated to patrons of the cooperative or beneficiaries of the estate or trust (see instructions)		**5**	
6	Cooperatives, estates, and trusts, subtract line 5 from line 4. Report this amount on Form 3800, Part III, line 4b		**6**	

For Paperwork Reduction Act Notice, see separate instructions. Cat. No. 13570D Form **5884** (Rev. 3-2021)

Walmart

Back

WOTC Questionnaire



Please take a few moments to answer the following questions from the Work Opportunity Tax Credit Program (WOTC).

Please note that -

- The questions take less than 3 minutes to complete
- Your responses are voluntary and may assist members of targeted groups in securing employment
- Your answers will be kept strictly confidential and will not adversely affect your employment opportunity

Please read the following disclaimer:

Once you start the WOTC questionnaire the set language cannot be changed until finished.

Let's get started!

At this time, please answer Yes or No to the following questions:

In the last year, have you or anyone you've lived with received SNAP (Supplemental Nutrition Assistance Program also referred to as food stamps)?

- ◯ Yes
- ◯ No

In the last two years, have you or anyone you've lived with received TANF (Temporary Assistance for Needy Families also referred to as welfare)?

- ◯ Yes
- ◯ No

Are you a veteran of the U.S. Military/Armed Forces?

- ◯ Yes
- ◯ No

Are you a person who has a disability?

- ◯ Yes
- ◯ No

Are you unemployed?

And

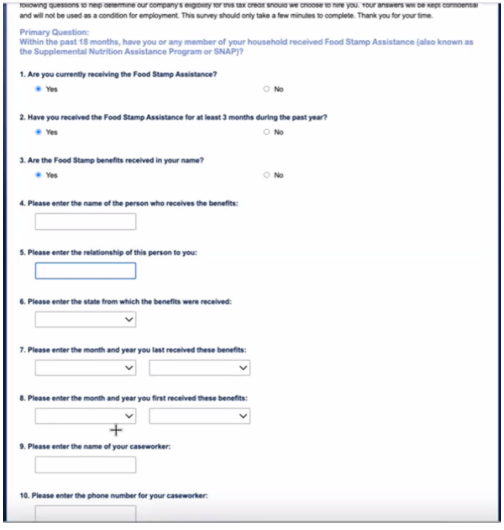
Have you collected unemployment benefits at any time during your unemployment period?

- ◯ Yes
- ◯ No

Are you or your spouse an enrolled member of a Native American tribe?

- ◯ Yes
- ◯ No

NSH Golden Age (nursing home) Back



following questions to help determine our company's eligibility for this tax credit should we choose to hire you. Your answers will be kept confidential and will not be used as a condition for employment. This survey should only take a few minutes to complete. Thank you for your time.

Primary Question:
Within the past 18 months, have you or any member of your household received Food Stamp Assistance (also known as the Supplemental Nutrition Assistance Program or SNAP)?

1. Are you currently receiving the Food Stamp Assistance?
- (•) Yes
- ( ) No

2. Have you received the Food Stamp Assistance for at least 3 months during the past year?
- (•) Yes
- ( ) No

3. Are the Food Stamp benefits received in your name?
- (•) Yes
- ( ) No

4. Please enter the name of the person who receives the benefits:

5. Please enter the relationship of this person to you:

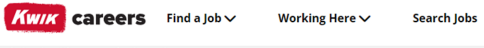
6. Please enter the state from which the benefits were received:

7. Please enter the month and year you last received these benefits:

8. Please enter the month and year you first received these benefits:

9. Please enter the name of your caseworker:

10. Please enter the phone number for your caseworker:



bring it with you to the interview.

We are an E-Verify Employer. Please click the links for more Information.
E-Verify Participation Information
IER Right to Work Information

Do any of the following statements apply to you?

• I am a member of a family that received (SNAP/Food Stamps) for 3 months or more during the past 6 months.

• I have received Supplemental Security Income (SSI) benefits for any month ending during the past 60 days.

• I have participated in a Vocational Rehabilitation Program or Ticket to Work Program.

• I've been unemployed for 6 months or more & received Unemployment Compensation at least once during that period.

• My family received Temporary Assistance For Needy Families (TANF) for 9 months or more during the past 2 years.

• I am a Veteran, AND;

- I am entitled to compensation for a service-connected disability. OR

- I am a member of a family that received SNAP/Food Stamps for at least 3 months during the past 15 months. OR

- I have been unemployed for more than 4 weeks during the past year.

Please Select

VOLUNTARY SELF IDENTIFICATION

The Equal Employment Opportunity Commission (EEOC) requires organizations with 100 or more employees to complete an EEO-1 Report each year. Completion

National Express

Back background Back mechanisms

WOTC Screening

Employment Tax Credit Eligibility

The information you have entered so far on your application has been saved.

National Express participates in the federal government's Work Opportunity Tax Credit (WOTC) program. This program was designed by the federal government to help companies with federal incentives for hiring and retaining individuals from targeted groups into the workforce.

We thank you in advance for completing the following online questionnaire. The information will screen for relevant company eligible tax credits should you be hired for this position. **Any information you provide will be kept confidential and will not negatively affect your job, wages, or taxes.**

Completing the questionnaire normally takes 2 to 4 minutes.

To start the questionnaire, click below.

Complete Questionnaire

EY Tax Credit Survey

Explanation of Survey

This employer participates in Federal and State tax credit programs such as the Work Opportunity Tax Credit (WOTC). These tax credit programs are offered by the federal government to increase hiring and improve retention of employees.

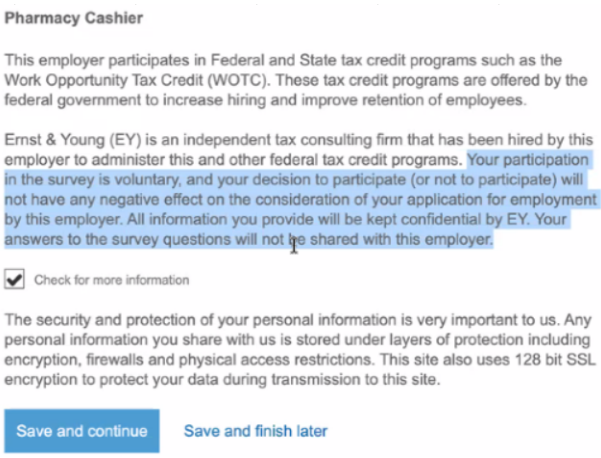
Ernst & Young (EY) is an independent tax consulting firm that has been hired by this employer to administer WOTC and other federal tax credit programs. As part of its administration of the tax credit program, EY requests that you complete the following on-line survey. Your participation in the survey is voluntary, but encouraged, and your decision to participate will not have any negative effect on the consideration of your application for employment by this employer. The information you provide will be used solely by EY to determine your eligibility under the tax credit program and this employer's eligibility to obtain tax credits. EY's determinations as to tax credit eligibility will not affect the consideration of your application for employment by this employer. All information you provide will be kept confidential by EY. Your answers to the survey questions will not be shared with this employer. *Click for more information.*

The security and protection of your personal information is very important to us. Any personal information you share with us is stored under layers of protection including encryption, firewalls and physical access restrictions. This site also uses 128 bit SSL encryption to protect your data during transmission to this site.

Your responses to the EY WOTC Survey will be handled in accordance with the EY WOTC Survey Privacy Notice.

Example

Back background | Back mechanisms



Pharmacy Cashier

This employer participates in Federal and State tax credit programs such as the Work Opportunity Tax Credit (WOTC). These tax credit programs are offered by the federal government to increase hiring and improve retention of employees.

Ernst & Young (EY) is an independent tax consulting firm that has been hired by this employer to administer this and other federal tax credit programs. Your participation in the survey is voluntary, and your decision to participate (or not to participate) will not have any negative effect on the consideration of your application for employment by this employer. All information you provide will be kept confidential by EY. Your answers to the survey questions will not be shared with this employer.

☑ Check for more information

The security and protection of your personal information is very important to us. Any personal information you share with us is stored under layers of protection including encryption, firewalls and physical access restrictions. This site also uses 128 bit SSL encryption to protect your data during transmission to this site.

Save and continue Save and finish later

Interaction with Ban the Box (BtB) policies

One of the groups subsidized by WOTC is individuals with felony convictions

- Could BtB policies limit WOTC efficacy by precluding questions on convictions?

Interaction with Ban the Box (BtB) policies

One of the groups subsidized by WOTC is individuals with felony convictions

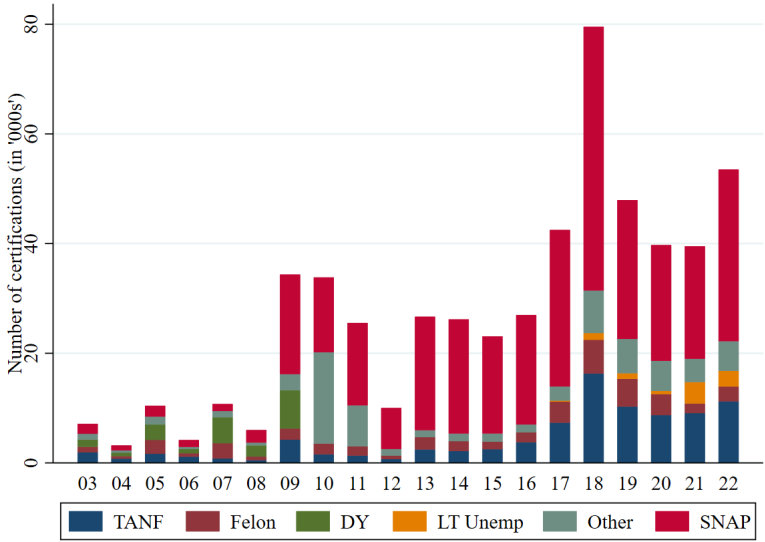
- Could BtB policies limit WOTC efficacy by precluding questions on convictions?

Probably not

1. In Wisconsin, BtB only applies to public sector employment
2. Employers can (and often do) ask about if employee is member of **any** WOTC-eligible group rather than about each category (EEOC-CVG-2020-1)
3. Applications can also ask about all the categories aside from felony conviction (>96% of WOTC applications)
4. Ban the Box doesn't preclude learning about convictions, but just doing so on the initial job application (Rose, 2021)

Back

Trends in WOTC claiming, Wisconsin back



Why are WOTC applications rejected?

Don't see reasons for rejection in our Wisconsin data

However, we have FOIAed data on WOTC applications for 7 other states, some of which document reasons for rejection. In North Dakota, the biggest reasons are:

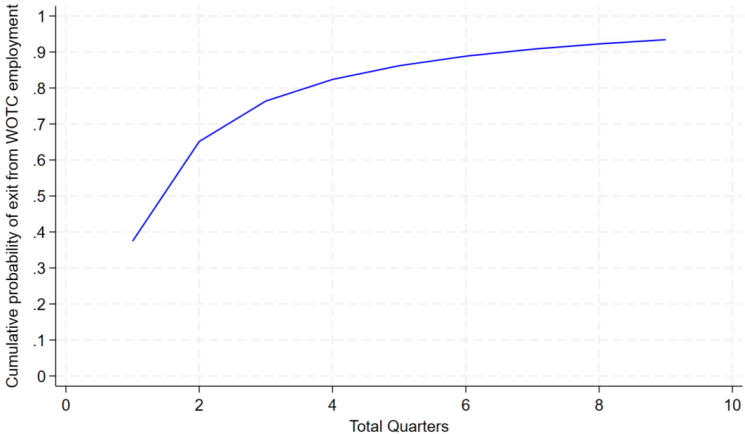
- 40%: not a member of target group
- 22%: did not submit supplementary documents within required time (90-120 days)
- 14%: part of target group, but did not qualify (e.g. age, time receiving benefits)
- 2%: previously worked for employer
- 2%: fail to submit within 28 days

back

WOTC-subsidized jobs are short-duration jobs

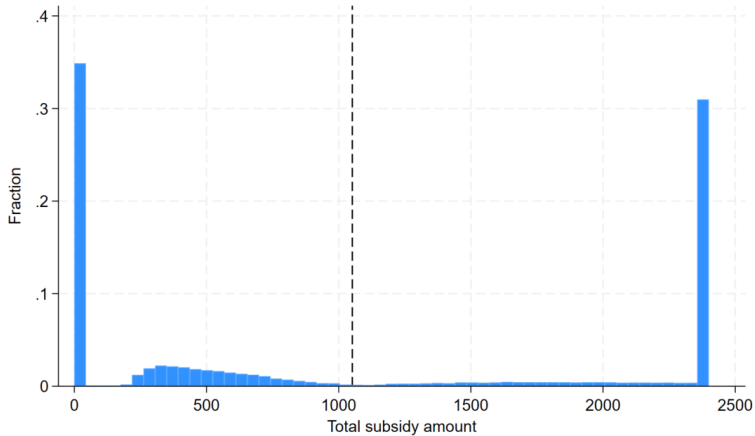
Back

Median duration is only 1-2 quarters, though approx. 10% remain for 2 or more years



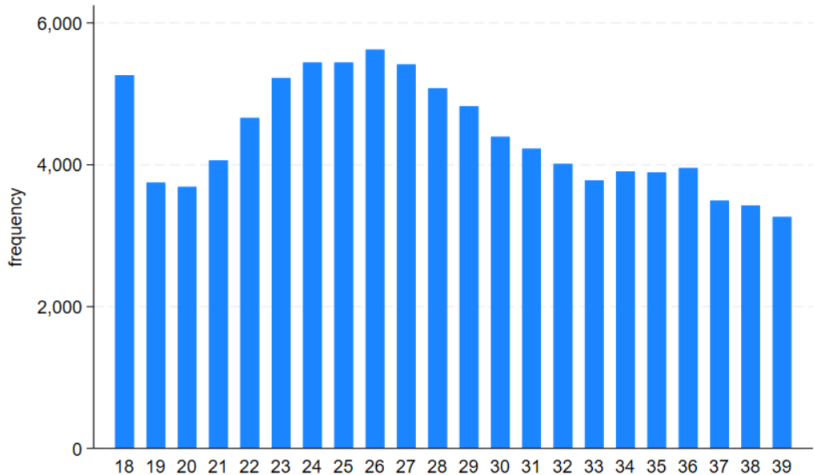
Average subsidy is $1,000; 1/3 get full subsidy

Back



Eligible population through SNAP in 2007Q1, by age

Back



Eligibility for SNAP

Gross income limit for SNAP in Wisconsin (called FoodShare) varies by household size, is 200% of poverty line back

- In 2010, this was $36,620 annually for a family of 3; now is $51,648
- Median wage in our data is $9 per hour
- Two adults working full-time at this wage would still qualify for SNAP

Since 1996, federal law placed limit on SNAP eligibility for able-bodied adults without dependents (3 months in a 36 month period)

- However, Wisconsin had this requirement waived starting in 2002 due to high unemployment; went back into effect as of April 1, 2015
- Numerous exemptions related to caregiving that many of our sample qualify for

Benchmark: what would standard models predict? Back

Implied subsidy rate at hiring: **19.2%**

- Total subsidy payments divided by total earnings over first $6,000 of wages, across all WOTC applications
- Would be higher if instead restricted to WOTC certifications

Apply standard formulas (Hamermesh, 1993) with conservative elasticities

- Labor demand elasticity of -0.1 to -0.2 (Lichter et al., 2015; Harasztosi and Lindner, 2019)
- Labor supply elasticity of 0.25 to 0.5 (Meyer, 2002; Chetty et al., 2013)
- $\Rightarrow$ predicted employment gains of **1.4-2.8%**

Larger elasticities imply larger predicted effects, so this is a lower bound; our estimates easily rule out effects of this size

Placebo RD

The key identifying assumption is that other employment-related factors don't change around age 40 or age 25

- Test this by looking at the age 40 discontinuity in the pre-2007 sample and age 25 discontinuity in the post-2007 sample
- Does not appear to be a pre-existing relationship between the labor market outcomes and the (combined) discontinuity

Placebo RD

The key identifying assumption is that other employment-related factors don't change around age 40 or age 25

- Test this by looking at the age 40 discontinuity in the pre-2007 sample and age 25 discontinuity in the post-2007 sample
- Does not appear to be a pre-existing relationship between the labor market outcomes and the (combined) discontinuity

	(1) WOTC certified job	(2) WOTC certified hire	(3) Employed	(4) New hire	(5) Earnings	(6) Public benefit (amt)	(7) Convicted
RD_Estimate	-0.0016	-0.0006	-0.0020	0.0001	-10.4453	-1.6621	0.0002
	(0.0010)	(0.0004)	(0.0018)	(0.0010)	(10.2166)	(3.6804)	(0.0003)
Dep var mean	0.0665	0.0257	0.5435	0.1557	2191.3618	1028.8860	0.0149
Bandwidth	2.7186	3.3757	2.6191	2.5547	2.5506	2.5347	3.7565
Effective Obs	2764889	3412276	3040328	2946658	2946658	2946658	3981643

return

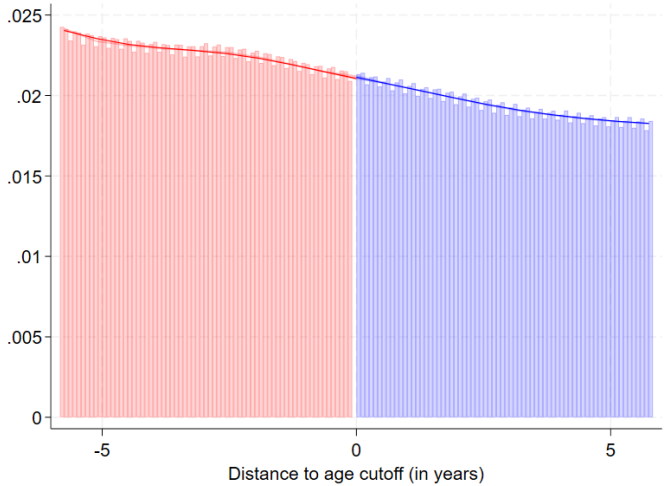
Compositional effects back

We find no evidence that the likelihood of receiving SNAP changes with WOTC eligibility, using both the 2007 expansion and the age discontinuities

	Approach 1: 2007 SNAP Expansion		Approach 2: Regression Discontinuity	
	Age 25 cut-off	Age 40 cut-off	Pre-2007	Post-2007
Treatment X Post	0.00107	0.00112	–	–
	(0.00082)	(0.00077)		
RD_Estimate	–	–	-0.00015	0.00025
			(0.00033)	(0.00043)
Dep var mean	0.0707	0.0553	0.0509	0.0960
Bandwidth			2.9596	3.7248
Effective Obs	5738040	4391950	21496548	30303913

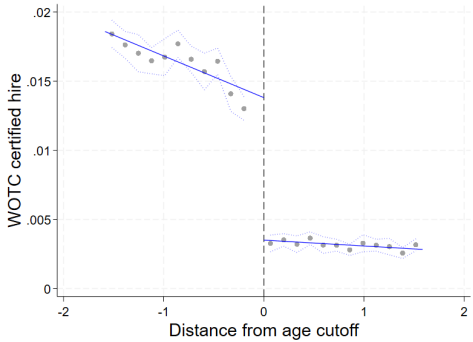
Tests for density of running variable back

No evidence of manipulation of age or differences in sample composition across the threshold as a result of WOTC (density test pooling both cutoffs, $p = 0.27$)



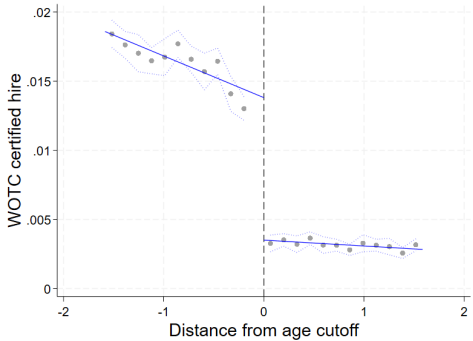
RD First stage - certifications

Difference of 1.0pp across the threshold in being hired into a WOTC-certified job



RD First stage - certifications

Difference of 1.0pp across the threshold in being hired into a WOTC-certified job



- Ruled out >0.27 percentage point increase in hiring at a 95% confidence level
- No more than 28% of hires were due to WOTC; at least 72% are inframarginal

back

RD analysis by firm type

back

No effect on being hired at specific types of firms that are heavier WOTC users

	Top 50 firm	Highly attentive firm	Explicitly screen	Staffing firm	High comp industry
Panel A: Employed					
RD_Estimate	0.0000	0.0004	0.0002	0.0003	-0.0013
	(0.0009)	(0.0007)	(0.0005)	(0.0009)	(0.0016)
Dep var mean	0.064	0.035	0.016	0.070	0.217
Bandwidth	3.686	3.576	4.120	3.832	4.002
Effective Obs	3601745	3438705	4010680	3684042	3929298
Panel B: New hire					
	Top 50 firm	Highly attentive firm	Explicitly screen	Staffing firm	High comp industry
RD_Estimate	0.0002	0.0001	0.0001	0.0002	0.0001
	(0.0004)	(0.0003)	(0.0001)	(0.0005)	(0.0006)
Dep var mean	0.018	0.010	0.003	0.028	0.051
Bandwidth	2.296	2.609	3.301	3.168	3.646
Effective Obs	2204601	2535660	3193002	3110432	3520309

RD analysis by individual characteristics

back

Table: Effect of WOTC on employment and hiring by individual characteristics

	Race		Gender		Past conviction	
	Black	Non-black	Female	Male	Yes	No
Panel A: Employed						
RD_Estimate	0.0001	-0.0011	0.0013	-0.0020	0.0012	0.0001
	(0.0035)	(0.0023)	(0.0025)	(0.0030)	(0.0042)	(0.0022)
Dep var mean	0.465	0.465	0.500	0.394	0.356	0.492
Bandwidth	3.653	3.728	3.035	5.359	4.284	2.843
Effective Obs	1,077,393	2,490,163	1,953,968	1,793,276	756,916	2,284,982
Panel B: New hire						
	Race		Gender		Past conviction	
	Black	Non-black	Female	Male	Yes	No
RD_Estimate	-0.0019	-0.0001	-0.0001	-0.0017	-0.0022	-0.0005
	(0.0019)	(0.0012)	(0.0013)	(0.0017)	(0.0020)	(0.0011)
Dep var mean	0.145	0.114	0.131	0.110	0.115	0.125
Bandwidth	3.068	2.726	2.777	2.858	4.113	2.826

RD analysis by year

back back to RD results

Re-estimate the RD within six four-year windows

- No evidence of differential effects over time as the program expands

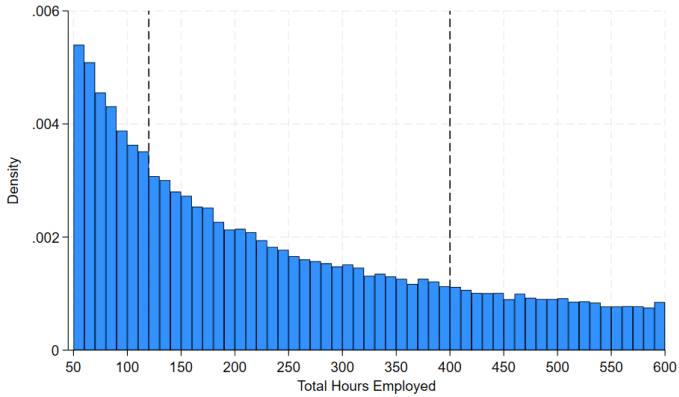




Subsidy notches: total hours and earnings in WOTC jobs back churning

Calculate total hours/earnings for each WOTC job, plot PDF

If firms terminate workers right after maximizing subsidy, will see bunching right after $6,000 in earnings and/or increase in slope right after; similar at 120/400 hours cdf



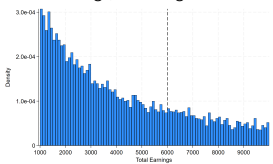
No churning or hazard response among firms most attentive to WOTC

back

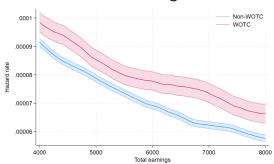
Also no evidence of bunching or hazard changes at top 50 WOTC users, highly attentive firms, staffing firms, highly competitive industries, or in more recent years

hazard model

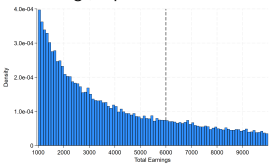
Earnings, screening firms



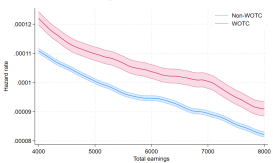
Hazard, screening firms



Earnings, top 50 WOTC firms



Hazard, top 50 WOTC firms



Hazard model of separation at subsidy exhaustion

back

Compare hazard of separation for WOTC hires vs. non-WOTC hires at WOTC-using firms, around $6,000 in earnings (difference-in-discontinuities in the hazard)

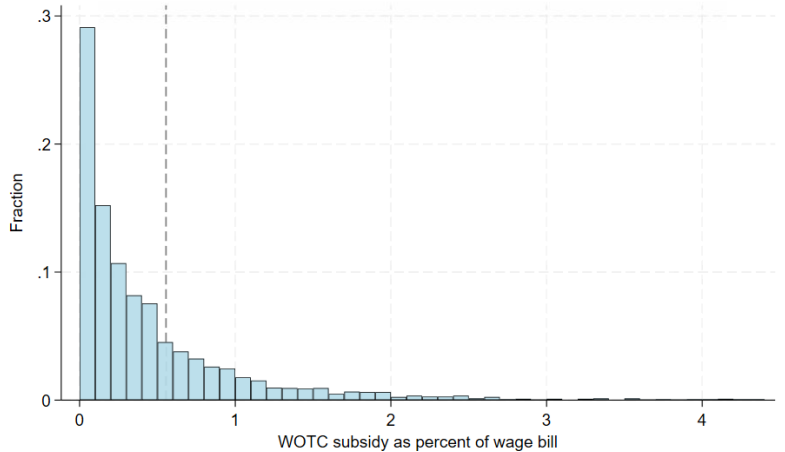
	(1) Hazard model	(2) Hazard model	(3) Hazard model
Post X WOTC	-0.0277	-0.0222	-0.0142
	(0.0244)	(0.0173)	(0.0122)
Earnings	-0.0027***	-0.0014***	-0.0008***
	(0.0000)	(0.0000)	(0.0000)
Post	1.4441***	1.3310***	1.2850***
	(0.0109)	(0.0077)	(0.0055)
Post X Earnings	-0.0024***	-0.0008***	-0.0002***
	(0.0000)	(0.0000)	(0.0000)
WOTC	0.0090	0.0040	-0.0108
	(0.0195)	(0.0135)	(0.0091)
Earnings X WOTC X Post	0.0001**	0.0001***	0.0000***
	(0.0000)	(0.0000)	(0.0000)
Earnings X WOTC	0.0000	0.0000	-0.0000*
	(0.0000)	(0.0000)	(0.0000)
Lower bound	5000	4000	2000
Upper bound	7000	8000	10000
Observations	696590	779176	885725

By firm type ($4,000-$8,000 window):

	(1) Top 50 firm	(2) Highly attentive firm	(3) Explicitly screen	(4) Staffing firm	(5) High comp industry
Post X WOTC	-0.0094	-0.0260	-0.0284	-0.0274	-0.0109
	(0.0261)	(0.0258)	(0.0549)	(0.0319)	(0.0360)
Lower bound	4000	4000	4000	4000	4000
Upper bound	8000	8000	8000	8000	8000
Observations	377887	444585	104673	260944	250320

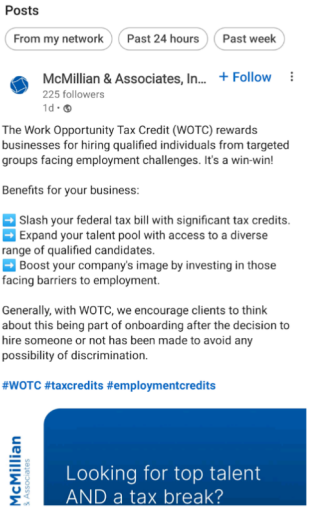
What share of payroll are WOTC subsidies? back

- Among firms that use WOTC, subsidy amount averages 0.55% of payroll (2015-6)
- This is similar to the state Unemployment Insurance Tax as a share of payroll



Onboarding example: tax specialist

Back



EEOC formal opinion letter (2020) back



U.S. Equal Employment Opportunity Commission

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Home » laws » guidance » Commission Opinion Letter: Federal Work Opportunity Tax Credit Form 8850

Commission Opinion Letter: Federal Work Opportunity Tax Credit Form 8850

April 29, 2020

[address]

Dear [redacted]:

This opinion letter responds to your recent inquiry about the Internal Revenue Service (IRS) Form 8850 (Form 8850) used in the Federal Work Opportunity Tax Credit (WOTC) programs. You asked whether the U.S. Equal Employment Opportunity Commission's (EEOC's) position regarding the use of Form 8850 has changed since our last correspondence with [redacted] in April 2009. Our position has not changed, and we continue to advise that proper employer use of Form 8850 would not violate the federal equal employment opportunity (EEO) laws, most prominently Title I of the Americans with Disabilities Act.

This guidance document was issued upon approval by vote of the U.S. Equal Employment Opportunity Commission.

OLC Control Number:	EEOC-CVG-2020-1
Concise Display Name:	Commission Opinion Letter: Federal Work Opportunity Tax

Firm-level matched DiD around 2007 expansion back

Worker-level designs could miss some firm responses

- Subsidy as cash injection $\Rightarrow$ firm-wide increases in hiring or wages
- Decision-makers may not exactly follow age rules $\Rightarrow$ screen "probabilistically" on correlates

Design: firms using WOTC before 2007 vs. propensity-score matched non-users

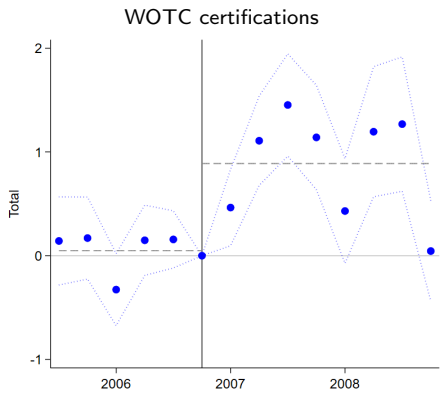
- Expansion created new incentive for WOTC users to hire SNAP recipients 25-39
- Match on pre-period share of hires who are SNAP 25-39, industry WOTC use, wage bill, quarterly hires balance

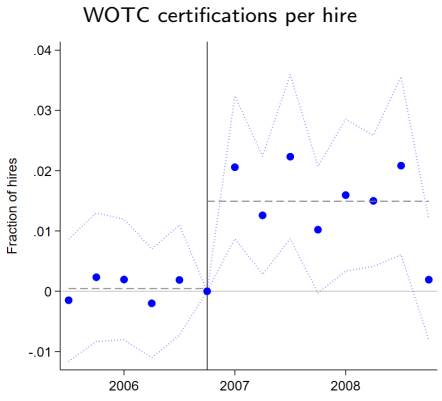
$$Y_{f,t} = \alpha_f + \delta_t + \beta\,(\text{Treated}_f \times \text{Post}_t) + \varepsilon_{f,t}$$

first stage hiring and payroll firm adoption design

Matched DiD: WOTC users receive more subsidies after expansion

back

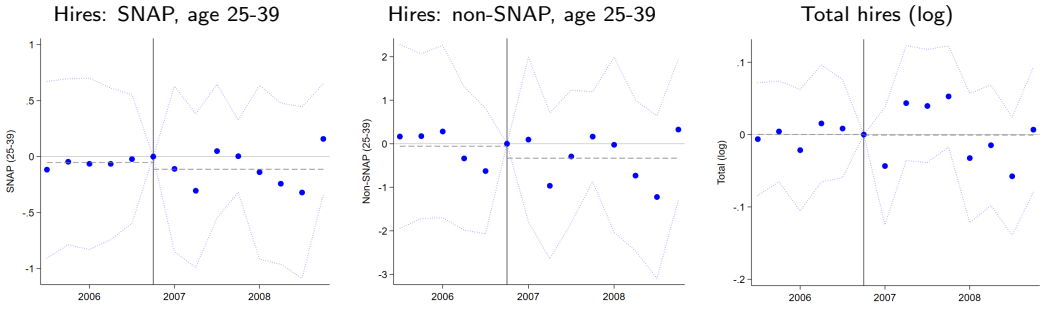




- Share of hires certified rises from 3.4% to 5.1%

Matched DiD: no change in hiring or payroll

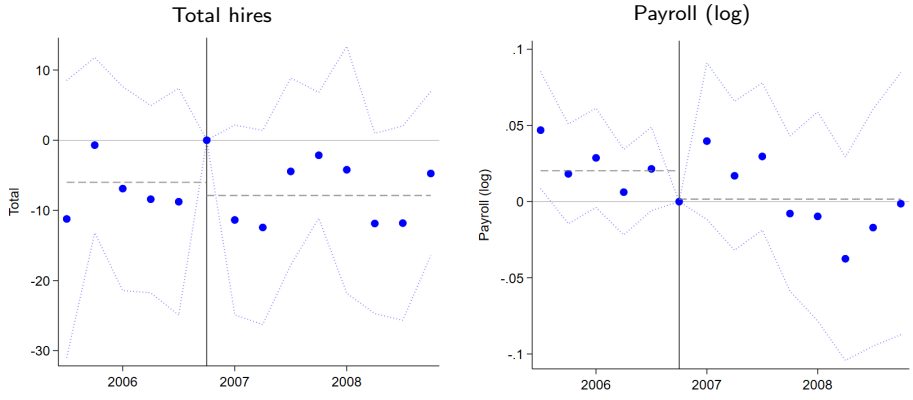
back



	WOTC certifications		Hires				Payroll
	Total	Fraction of hires	SNAP (25-39)	Non-SNAP (25-39)	Total	Total (log)	Payroll (log)
Treat x post	1.071***	0.017***	-0.064	-0.227	-0.715	0.005	-0.020
	(0.194)	(0.003)	(0.143)	(0.392)	(2.658)	(0.023)	(0.027)
Dep var mean	2.1362	0.0341	2.7851	8.2130	75.2400	3.3210	6.9735
Observations	40720	39229	40720	40720	40720	39229	39836

total hires, payroll firm adoption design

Matched DiD: total hires and payroll back



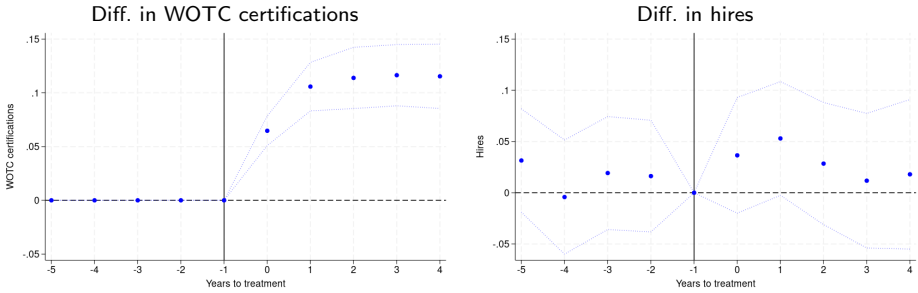
Subsidies are a small share of payroll given low wages, so too small to generate firm-wide adjustments payroll share

Matched DiD: balance between treated and matched firms back

Variable	Treated mean	Untreated mean	Difference	p-value
Avg quarterly wage bill (ln)	6.992	6.987	.004	.961
			(.082)	
Avg quarterly hires (ln)	3.465	3.467	-.002	.979
			(.076)	
Share hires: SNAP recipients 25-39	.038	.038	0	1
			(.002)	
Avg quarterly workers	516.306	663.491	-147.185	.285
			(137.709)	
Average WOTC usage (NAICS 4 digit)	.037	.035	.002	.617
			(.004)	

Firm-level adoption design matched DiD back

- Event study around firm's first WOTC application vs. not-yet-adopters (Callaway and Sant'Anna, 2021)
- Outcome: difference in certifications/hires of just-eligible vs. just-ineligible SNAP recipients (38-39 vs. 40-41)



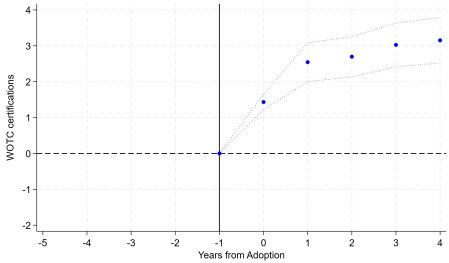
Sharp, sustained increase in certifications but no change in relative hiring table

Firm-level adoption design: estimates

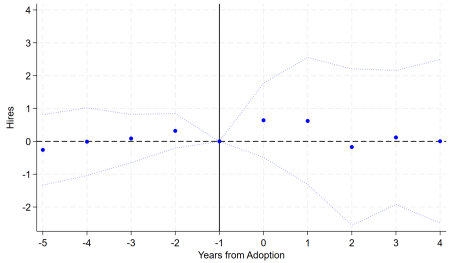
back

Also no shift toward SNAP recipients overall (SNAP vs. non-SNAP hires aged 18-39)

Diff. in WOTC certs, SNAP vs. non-SNAP 18-39



Diff. in hires, SNAP vs. non-SNAP 18-39



	38–39 SNAP vs 40–41 SNAP		35–39 SNAP vs 40–44 SNAP		18–39 SNAP vs 18–39 Non-SNAP	
	Diff. in WOTC certs.	Diff. in hires	Diff. in WOTC certs.	Diff. in hires	Diff. in WOTC certs.	Diff. in hires
ATT	0.100***	0.024	0.310***	0.026	2.460***	0.184
	(0.008)	(0.020)	(0.023)	(0.041)	(0.200)	(0.575)
Observations	51884	51884	51884	51884	51884	51884

Triple-differences: adding non-SNAP recipients of same ages

back

Difference out age-specific labor market shocks using non-SNAP recipients in the same age ranges

	WOTC certified job	WOTC certified hire	Employed	New hire	Earnings	Public benefit (amt)	Convicted
Panel A: Stacked sample							
Treatment X Post X SNAP	0.0203*** (0.0015)	0.0111*** (0.0007)	0.0008 (0.0040)	-0.0014 (0.0023)	-36.6606 (23.4531)	9.5545 (8.1312)	0.0001 (0.0007)
Dep var mean	0.0038	0.0016	0.4209	0.0664	3509.6788	94.1580	0.0052
Observations	10659165	10659165	10659165	10659165	10659165	10659165	10659165
Panel B: Age 40 cutoff sample							
	WOTC certified job	WOTC certified hire	Employed	New hire	Earnings	Public benefit (amt)	Convicted
Treatment X Post X SNAP	0.0142*** (0.0012)	0.0070*** (0.0006)	0.0033 (0.0064)	-0.0033 (0.0032)	-19.2365 (38.9758)	28.2710** (13.1319)	0.0000 (0.0012)
Dep var mean	0.0007	0.0003	0.4305	0.0422	4782.5802	90.3545	0.0036
Observations	4724736	4724736	4724736	4724736	4724736	4724736	4724736
Panel C: Age 25 cutoff sample							
	WOTC certified job	WOTC certified hire	Employed	New hire	Earnings	Public benefit (amt)	Convicted
Treatment X Post X SNAP	0.0241*** (0.0023)	0.0137*** (0.0011)	-0.0039 (0.0051)	-0.0006 (0.0031)	-45.1180 (29.0934)	-2.6182 (10.3803)	0.0001 (0.0010)
Dep var mean	0.0063	0.0027	0.4132	0.0857	2497.7940	97.1816	0.0065
Observations	5934429	5934429	5934429	5934429	5934429	5934429	5934429

DD analysis by firm type back

No effect on being employed or hired at firm types most likely to respond

	Top 50 firm	Highly attentive firm	Explicitly screen	Staffing firm	High comp industry
Panel A: Employed					
Treat X Post	-0.0018	0.0015	-0.0017	-0.0000	0.0048
	(0.0023)	(0.0019)	(0.0014)	(0.0021)	(0.0037)
Dep var mean	0.0846	0.0558	0.0258	0.0775	0.2491
Observations	527241	527241	527241	527241	527241
Panel B: New hire					
	Top 50 firm	Highly attentive firm	Explicitly screen	Staffing firm	High comp industry
Treat X Post	0.0011	0.0000	-0.0001	0.0012	-0.0013
	(0.0009)	(0.0008)	(0.0004)	(0.0011)	(0.0015)
Dep var mean	0.0236	0.0180	0.0044	0.0336	0.0652
Observations	527241	527241	527241	527241	527241

RD balance on pre-determined characteristics back

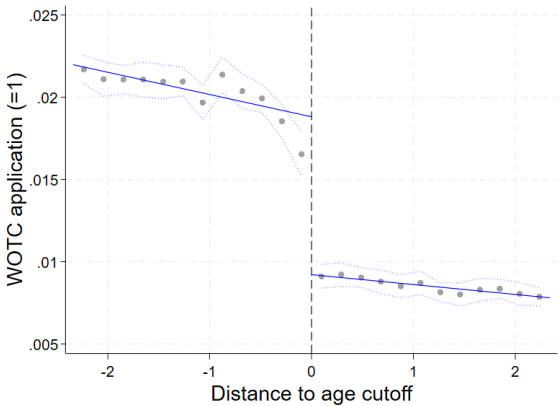
No discontinuities in gender, race, or lagged employment and earnings (pooling pre- and post-2007)

	Demographics		Employed		Earnings	
	Female (=1)	Black (=1)	(t-1)	(t-2)	(t-1)	(t-2)
RD_Estimate	0.0012	-0.0004	-0.0006	0.0002	3.8123	7.8239
	(0.0021)	(0.0021)	(0.0020)	(0.0020)	(11.7246)	(11.3883)
Dep var mean	0.6540	0.3054	0.4833	0.4779	1972.7684	1857.3763
Bandwidth	2.5335	4.1067	3.2818	3.0561	4.3648	4.2236
Effective Obs	2453543	4000381	3045416	2794327	4057170	3878363

Do firms know the age cutoff? Yes back

WOTC application rates jump at the age eligibility threshold

- Someone at the firm knows the age rule, but this information may not reach hiring decision-makers



Initial job applications for a sample of WOTC-using firms

back

Panel A: All Applications (N=138)	
Characteristic	(% of applications)
WOTC Screening Questions	21.7
Submit Resume	70.4
Education History	56.6
Past Employment History	69.5
References	11.1
Panel B: Applications with WOTC Screening Questions (N=30)	
Characteristic	(% of applications)
Third party firm handles WOTC data	60.0
WOTC questions state they have no effect on hiring	30.0
WOTC questions state they have no negative impact on hiring	46.7

Firms drawn with probability proportional to WOTC application volume, so statistics are WOTC-application-weighted

Up-front costs and attentiveness: screening firms

back

Firms that screen for WOTC in-house **and** (a) are top 50 WOTC users or (b) are highly attentive to WOTC

- Top 50 firms receive >$165,000 per year in WOTC subsidies on average
- Highly attentive screening firms are certified for 48.7% of potentially eligible hires (vs. 24.6% at other firms)

Panel A: DiD estimates on labor market outcomes at screening firms

	Top 50 firm		High attentiveness firm	
	New hire	Employed	New hire	Employed
Treat X Post	-0.0000	-0.0008	-0.0002	-0.0013
	(0.0004)	(0.0013)	(0.0003)	(0.0012)
Dep var mean	0.0038	0.0201	0.0027	0.0164
Observations	504104	504104	504104	504104

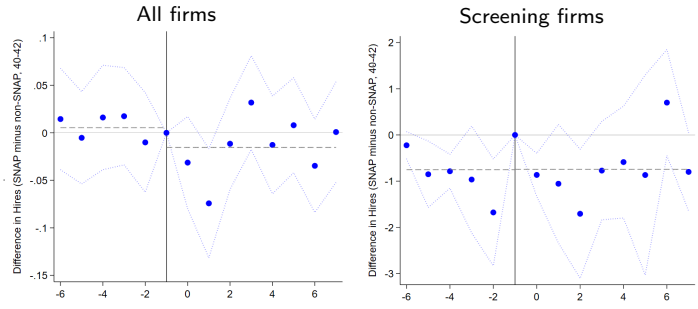
Panel B: RDD estimates on labor market outcomes at screening firms

	Top 50 firm		High attentiveness firm	
	New hire	Employed	New hire	Employed
RD_Estimate	0.0001	-0.0000	0.0001	-0.0001
	(0.0001)	(0.0004)	(0.0001)	(0.0004)
Dep var mean	0.001	0.010	0.002	0.011
Bandwidth	4.586	4.910	4.130	4.693
Effective Obs	4316680	4550985	3847469	4394464

Stigma: does observing SNAP receipt reduce hiring? back

After adopting WOTC, firms learn SNAP status of applicants, including those aged 40+ who bring no subsidy

- If SNAP receipt is a negative signal, adopters should hire fewer SNAP recipients aged 40-42 (vs. non-SNAP 40-42)
- Matched stacked DiD around adoption: no change, including at firms that screen for WOTC



e-filing test

Are eligible and ineligible workers substitutable? back

Standard model can only rationalize null effects with perfectly inelastic demand for eligible workers

- Test using hiring responses to worker exits (Jäger and Heining, 2022)
- If not substitutable, exit of eligible worker (SNAP, 37.75-39.75) should be replaced only by another eligible worker
- Similar replacement of eligible exits by eligible (SNAP 37.75-39.75) and ineligible (SNAP 42-44) hires $\Rightarrow$ highly substitutable

	(1) Eligible hires	(2) Eligible hires	(3) Ineligible hires	(4) Ineligible hires
Eligible exits	0.124***	0.084***	0.107***	0.086***
	(0.025)	(0.013)	(0.020)	(0.011)
Ineligible exits	0.151***	0.114***	0.119***	0.087***
	(0.026)	(0.014)	(0.027)	(0.013)
Firm FEs	Yes	Yes	Yes	Yes
Year-Quarter FEs	Yes	Yes	Yes	Yes
Total Exits FEs	No	Yes	No	Yes
Observations	615668	615264	615668	615264

Do firms screen implicitly on correlates of eligibility? back

Re-purpose audit study of Kline, Rose and Walters (2024): 83,000+ applications to entry-level jobs, randomized age

- Link firm names to FOIAed WOTC data (2018-2020) to identify WOTC-using firms
- RD in callback rates around age 40, the WOTC eligibility cutoff for SNAP recipients
- No differential callbacks, including for applicants more likely to be seen as SNAP recipients

	Type of firm		Diff-in-RD	Applicant type (WOTC firms)	
	WOTC firm	Non-WOTC firm	.	Black	Non-college
RD_Estimate	-0.038	0.011	-0.049	-0.054	-0.026
	(0.029)	(0.038)	(0.048)	(0.034)	(0.032)
Dep var mean	0.241	0.242		0.241	0.242
Bandwidth	7.627	8.070		6.657	7.547
Effective Obs	19946	12755		8627	10227

Survey of supervisors of low-wage workers back

Prolific survey of individuals in leadership positions at for-profit firms that use WOTC

- Supervise low-wage workers; pass knowledge check on how WOTC works
- 62 respondents; compensated at average of $23/hour

What matters for termination decisions?

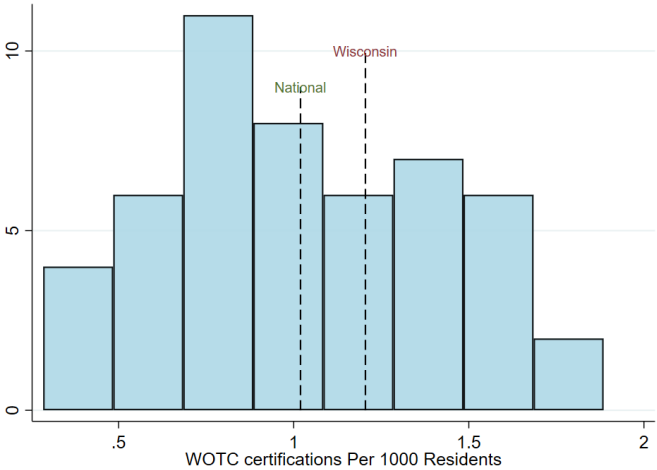
- Worker productivity (97% "very important")
- Business conditions (87% somewhat/very important)
- WOTC: 66% no role, 11% only a small role

Why doesn't WOTC factor into these decisions?

- 85%: managers do not directly benefit from the tax credit (only the company does)
- 71%: managers often do not know which employees are bringing WOTC money in
- 71%: credit is small relative to the cost of hiring and training a new worker

WOTC certifications per capita, by state back

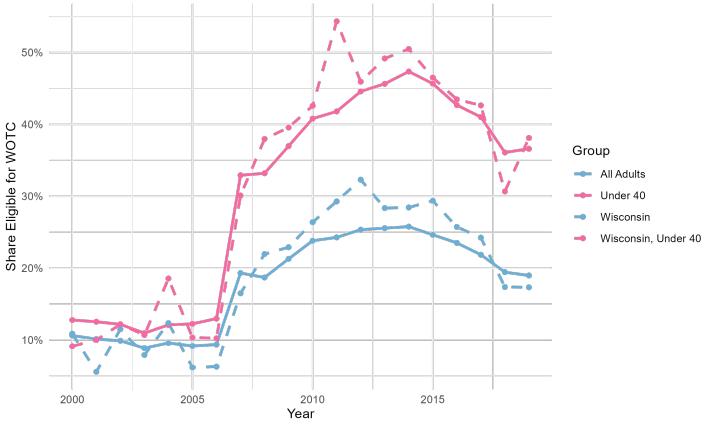
Wisconsin's per capita WOTC utilization closely tracks the national average



Share of individuals in poverty who would be WOTC-eligible back

CPS-based estimate of share below poverty line who would be WOTC-eligible if hired

- Jumps after 2007 expansion; 36-47% of poor adults aged 18-40 since 2010 (lower bound)

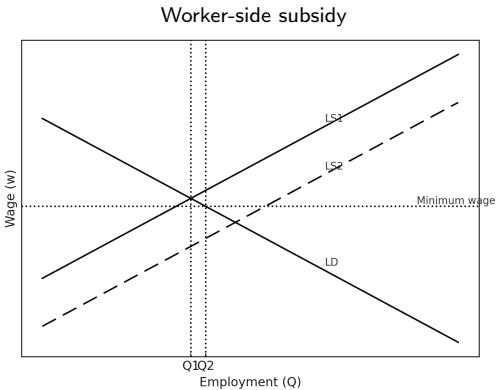


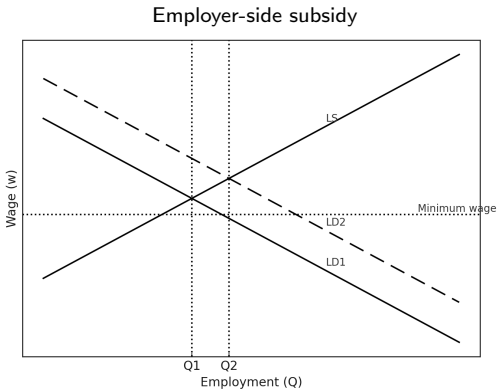
Worker-side vs. employer-side subsidies with a binding minimum wage

back

With a binding minimum wage, wages cannot fall in response to a worker-side subsidy, so employer-side subsidies have larger employment effects

- 58% of WOTC-subsidized workers earn within 25% of the $7.25 minimum wage





Negative signals and electronic filing

back e-filing results

Even if the expected value of the subsidy balances out the negative signal, we would still expect a response to electronic filing

- Suppose firms hire if expected benefit exceeds cost of applying:

$$E(Bt + S(t)) - pL > c$$

  where B is expected productivity, t is time worked, S(t) is subsidy, pL is expected loss, and c is cost of applying for WOTC

- Learning of WOTC status may decrease B and/or increase p in a way that balances out
- However a decrease in c, as under eWOTC, should unambiguously increase hiring

Job applications to WOTC firms back

- List of WOTC applications in WI in 2018-2020 with firm names (FOIAed data)
- Draw random sample of 179 firms with probability proportional to number of WOTC applications
- Find entry-level job applications for 77% of these (138 firms), jointly 53% of WI WOTC certifications
- Record whether application collects WOTC information, who collects it, and how its use is described
  - 22% ask WOTC questions; of these, 60% send it to a third party, 30% say it has no effect on hiring, 47% say it will not negatively affect hiring table
  - RA cross-check agrees in 97% of cases; practices persistent (14 of 15 firms unchanged since 2020-21)
- Add 60 more top WOTC filers to study firms that screen (coverage: 62.6% of WI WOTC applications)

Survey of hiring decision-makers back

Three rounds using Prolific, following Agan et al. (2024)

- US-based respondents aged 25+ with experience making hiring decisions
- Screen to those at firms that participate in WOTC
- Round 3: 99%+ approval rate and willing to record audio/video, to limit AI-bot responses; responses similar across rounds
- Analysis sample: 65 respondents whose firms receive WOTC but do not screen before hiring
- Compensated at average rate of $30/hour

Asked questions about:

- Their history in hiring and using WOTC
- How their firms screen workers for WOTC eligibility and file applications
- How WOTC eligibility affects hiring decisionmaking and why

Why don't **workers** respond to these incentives? back

Another question is why workers don't respond to this - can imagine using subsidy to demand a higher wage

A few likely explanations:

- Fixed wages at the firm level – for low-wage jobs at large firms, these are likely to be standardized rather than individually bargained
- Salience - program is designed to not be salient to workers, who mostly will not know if firm is receiving subsidies